

Network Foundations – Online Food Delivery Application

Task 1: TCP/IP Communication Flow

Explain how data travels when a customer places a food order using the application.

Explanation

1. User's Device:

The customer selects food items and clicks Place Order. The application creates an order request containing details such as the selected items, quantity, delivery address, and payment information.

2. Request Transmission:

The request is sent from the customer's device through the Internet toward the application's server. IP is responsible for addressing and routing the data packets to the correct destination.

3. TCP Communication:

TCP provides reliable and ordered delivery of the data between the customer's application and the server. It helps ensure that data is delivered correctly and retransmits data if necessary.

4. Server Processing:

The server receives the request, processes the order, and performs the required operations such as creating the order and updating its status.

5. Response Transmission:

After processing the request, the server sends a response back through the network. TCP provides reliable delivery, while IP routes the response back to the customer's device.

6. Result on User's Device:

The application receives the server's response and displays the result to the customer, such as "Order Confirmed" along with the order details.

**Customer's Device → Internet → Application Server → Order Processing
→ Response to Device**

Task 2: Choose the Right Communication Protocol

Select the most suitable protocol for each scenario

Scenario	Protocol	Reason
Customer logs into the application securely	HTTPS	HTTPS uses TLS to encrypt the communication and protect sensitive login information such as usernames and passwords.
Live delivery tracking	WebSocket	WebSocket provides continuous two-way communication, making it suitable for receiving real-time delivery location and status updates.
Communication between the Order Service and Payment Service	gRPC	gRPC is suitable for efficient service-to-service communication between backend services such as Order Service and Payment Service.
Viewing the restaurant menu	HTTP	HTTP is suitable for simple request-response communication where the application requests menu data and the server returns it.

Task 3: Network Performance

Customers notice that the delivery tracking updates are delayed by 4–5 seconds.

Common Causes of Network Latency

- Physical distance
- Network congestion
- Slow routing
- Server processing time

1. Is this mainly a latency or throughput issue?

This is mainly a **latency issue**. The delivery tracking updates are arriving 4–5 seconds late, which means there is a delay in communication between the user's device and the server.

2. Two simple ways to improve the user experience.

1. **Reduce network latency:** Use servers or services that are geographically closer to customers and optimize the network communication to reduce the time required for updates to travel.
2. **Optimize real-time updates:** Use an efficient real-time communication method such as **WebSocket** so that delivery updates can be sent immediately instead of repeatedly requesting updates from the server.

Task 4: Secure Communication

1. Why should customer payment information be sent using HTTPS (TLS)?

Customer payment information should be sent using HTTPS (TLS) because TLS encrypts the data while it travels between the customer's device and the server. This helps prevent attackers from reading or modifying sensitive information such as card details, passwords, and payment information.

2. Why might employees access the company's internal systems using a VPN?

Employees might use a VPN (Virtual Private Network) to securely connect to the company's internal network when working remotely. The VPN creates a protected connection over the Internet, helping prevent unauthorized users from accessing or intercepting the company's internal communication.

Task 5: Modern Application Architecture

The application has the following services:

- **Order Service**
- **Payment Service**
- **Delivery Service**

When an order is confirmed, the Delivery Service automatically assigns a delivery partner.

1. Is this an example of a distributed system?

Yes. It is a distributed system because the application consists of multiple services—**Order Service, Payment Service, and Delivery Service**—that work together and communicate over a network to provide the food delivery functionality.

2. Is assigning the delivery partner an example of event-driven architecture?

Yes, if the Delivery Service is triggered by an event such as OrderConfirmed. When the Order Service publishes the event, the Delivery Service receives it and automatically assigns a delivery partner, allowing the services to communicate asynchronously without requiring the Order Service to directly control the assignment.